ROBOTICS 4

TOPIC 5: DENAVIT-HARTENBERG KINEMATIC REPRESENTATION

5.1 DEGREES OF FREEDOM, DEGENERACY AND DEXTERITY

DEGREES OF FREEDOM: The *degree of freedom* for a robot is the number of different joint motions that the robot configuration allows. This is not dependent on the type of joint, just the different directions the joints can move. The degrees of freedom for the common types of robot manipulators are listed in Table 1. This does not include the degrees of freedom of the end effector.

Table 1: Robot Joints and Degrees of Freedom

Robot Type	Number of Revolute Joints	Number of Prismatic Joints	Degrees of Freedom
Cartesian/Rectangular	0	3	3
Cylindrical	1	2	3
SCARA	2	1	3
Spherical	2	1	3
Anthropomorphic	3	0	3

DEGENERACY: Degeneracy occurs when a robot loses a degree of freedom and cannot operate properly. This occurs under two conditions:

1. When the robot reaches the physical limit of one or more of its joints.
2. When the *z*-axis of similar joints become collinear.

In either case the robot's controller will be unable to make the robot move in the desired manner (particularly in the latter case where movement of any joint will result in the same overall motion). In such cases the controller must take action to either avoid these scenarios or take emergency action to get out of these degenerate situations.

DEXTERITY: Theoretically it is possible to position and orientate a six degree of freedom robot at any desired location. However, as the robot reaches the limits of the workspace it is found that it may be possible to position the end effector as desired but orientation becomes difficult to ensure. The points where the position of the robot can be ensured but the orientation cannot is called the *nondexterous volume*.

5.2 DENAVIT-HARTENBERG REPRESENTATION

The Denavit-Hartenberg (D-H) Representation is a commonly used method of representing the kinematics of all robot manipulators, irrespective of the joint type and configuration. It assumes that the robot is made up of a succession of joints and links, in any direction, shape and orientation. The kinematic model is constructed using the joint/link configuration. To do this a local reference frame is assigned to each and every joint and the transforms from one joint to the next define the kinematics [1].

The local reference frames are defined in terms of the *z* and *x* axes (the y-axis is not used as it is known that it is mutually perpendicular to the other two axes). The following procedure is used to define these reference frames [1]:

- All joints are represented by a z-axis. For revolute joints the *z*-axis defines the axis of rotation with the *right hand rule* used to determine the direction of rotation (i.e. the fingers in the right hand are formed into a fist and the thumb is protruded upwards, the fingers show the direction of rotation). The angle of rotation (θ) is the joint variable. With prismatic joints the *z*-axis lies along the linear direction of motion. The length of the link (d) is typically the joint variable.
- The joint *z*-axes may not be parallel or intersecting, and therefore are skew lines. There is always one line that is mutually perpendicular to any two skew lines, which is called the *common normal*. The *x*-axis is always assigned to lie in the direction of the common normal between the current considered joint and the previous joint. The

common normals between successive joints are not necessarily intersecting or collinear. Therefore the location of the origins of two successive joint frames may not be at the same position.

- If the z-axes of two successive joints are parallel there are an infinite number of common normals between the axes. The one that is selected is collinear with the common normal from the previous joint. This simplifies the model.
- If the z-axes intersect there is no common normal. The x-axis is defined as the line perpendicular to the plane formed by the two z-axes. This simplifies the model.

Once the reference frames have been defined, certain variables can be defined. There are four such variables: θ represents the rotation about the z-axis (revolute joint angle), d represents the distance on the z-axis between two successive common normals (prismatic joint length), a represents the length of each common normal (called the *joint offset*) and α represents the angle between two successive z-axes (called the *joint twist*). Usually only θ and d are joint variables, but this is not always the case.

The next step is to follow the necessary motions to transform from one joint reference frame to the next. Consider the generic joint/link configuration in Figure 5.1 [1].

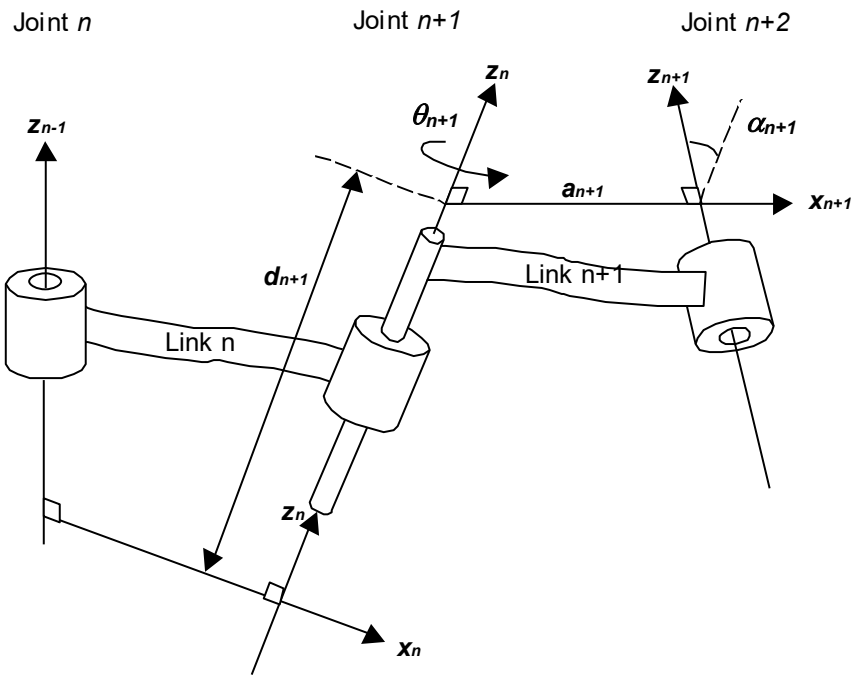
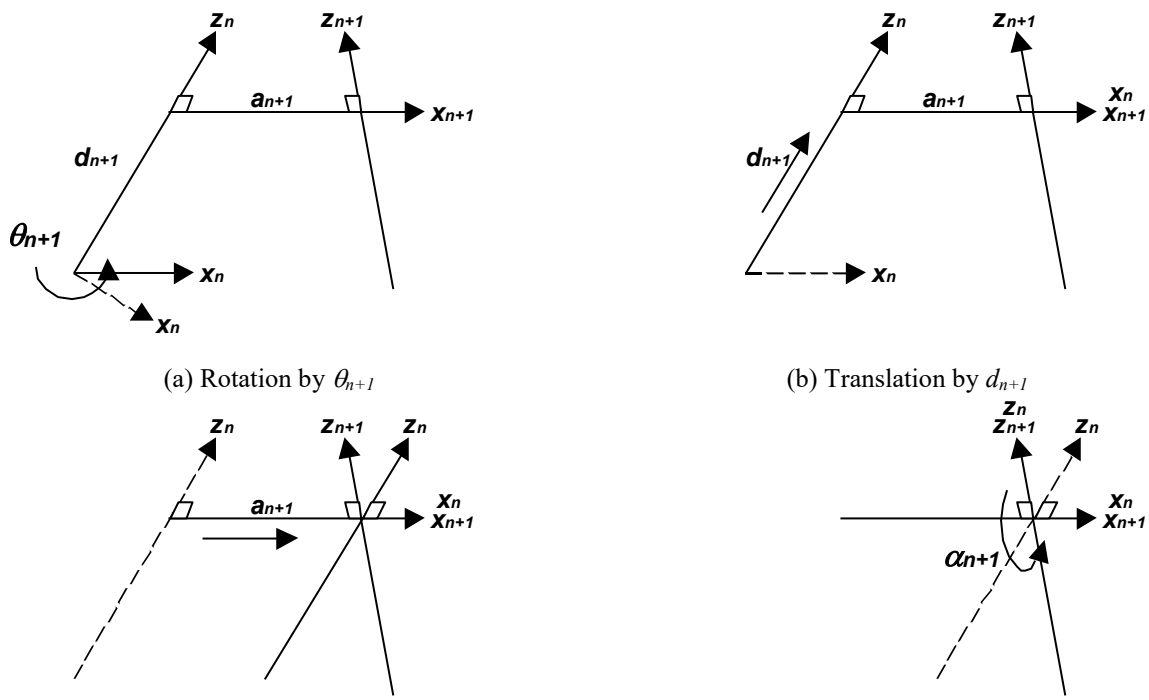


Figure 5.1: Generic Link Configuration

The analysis starts at the x_n - z_n reference frame. There are four standard motions to get from this reference frame to the next i.e. x_{n+1} - z_{n+1} reference frame. These are used to define the D-H representation thus:

- **Rotate about z_n-axis by angle θ_{n+1}** (see Figure 5.2(a)). This makes x_n and x_{n+1} parallel to each other.
- **Translate along the z_n-axis by a distance d_{n+1}** (see Figure 5.2(b)). This makes x_n and x_{n+1} collinear and lays one on top of the other.
- **Translate along the x_n-axis by a distance a_{n+1}** (see Figure 5.2(c)). This brings the origin of the two reference frames together.
- **Rotate z_n-axis about the x_{n+1}-axis by an angle α_{n+1}** (see Figure 5.2(d)). This aligns z_n-axis with z_{n+1}-axis and the transformation is complete.

In the D-H representation these four basic transformations can describe the overall transformation from the current joint reference frame to the next reference frame. This applies to any joint and its neighbour [1,2].



(a) Rotation by θ_{n+1} (b) Translation by d_{n+1}

(c) Translation by a_{n+1} (d) Rotation by α_{n+1}

Figure 5.2: Reference Frame Transformations

The transformation matrix for these four movements is found by post-multiplying the transformations of each separate motion i.e.

$$
\begin{aligned}
{}^{n}T_{n+1} &= A_{n+1} = Rot(z_n,\theta_{n+1}) \times Trans(0,0,d_{n+1}) \times Trans(a_{n+1},0,0) \times Rot(x_n,\alpha_{n+1}) \\
&= \begin{bmatrix} C\theta_{n+1} & -S\theta_{n+1} & 0 & 0 \\ S\theta_{n+1} & C\theta_{n+1} & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & d_{n+1} \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & 0 & a_{n+1} \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & C\alpha_{n+1} & -S\alpha_{n+1} & 0 \\ 0 & S\alpha_{n+1} & C\alpha_{n+1} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \\
&= \begin{bmatrix} C\theta_{n+1} & -S\theta_{n+1}C\alpha_{n+1} & S\theta_{n+1}S\alpha_{n+1} & a_{n+1}C\theta_{n+1} \\ S\theta_{n+1} & C\theta_{n+1}C\alpha_{n+1} & -C\theta_{n+1}S\alpha_{n+1} & a_{n+1}S\theta_{n+1} \\ 0 & S\alpha_{n+1} & C\alpha_{n+1} & d_{n+1} \\ 0 & 0 & 0 & 1 \end{bmatrix}
\end{aligned}
\tag{5.1}
$$

By starting at the base of the robot and considering the transformations for each joint from there to the hand of the robot, the total transformation kinematic relationship for the robot can be found i.e.

$$
{}^{U}T_H = {}^{U}T_1\,{}^{1}T_2\,{}^{2}T_3 \cdots {}^{m-1}T_m = A_1 A_2 A_3 \ldots A_m \tag{5.2}
$$

Where m is the total number of joints e.g. for a six degree of freedom robot with six joints, there will be six A transformation matrices [1,2].

To make the implementation of the D-H representation easier (particularly for multi-jointed robots), a table is constructed to detail the D-H variables for each joint. This table takes the form show in Figure 5.3 [1,2].

#	θ	d	a	α
1				
2				
3				
4				
5				
6				

Figure 5.3: D-H Parameter Table

By completing this table for each joint of the robot the parameters can be inserted into a separate A transformation matrix (Equation (5.1)) for each joint. The resulting transformation matrices are then post-multiplied as shown in Equation (5.2). This gives the total forward kinematic transformation matrix for the robot using the D-H representation.

5.3 Inverse Kinematic Solution of A Robot

The inverse kinematic solution is more important than the forward kinematics (as the inverse provides the joint information for a given end effector position and orientation. When the D-H representation is considered for multiple joints it can be seen that the resulting transformation matrix is very complex and involves a multitude of coupled angles. This makes it impossible to find enough individually accessible elements to find all the unknown joint variables. As with the previous coordinate transformations, the general approach to solving the inverse kinematics for a robot involves pre-multiplying by the inverse transformation matrices and solving for accessible variables at each stage [1] i.e.

$$A_1^{-1}\,{}^{U}T_H = A_1^{-1} A_1 A_2 A_3 \ldots A_m = A_2 A_3 \ldots A_m \tag{5.3}$$

Then solve for available variables and then pre-multiply again:

$$A_2^{-1} A_1^{-1}\,{}^{U}T_H = A_2^{-1} A_2 A_3 \ldots A_m = A_3 \ldots A_m \tag{5.4}$$

This continues until all the joint variables have been obtained. This is the general solution that was applied to the inverse kinematic solutions for the previous coordinate transformations.

5.4 Problem with Denavit-Hartenberg Kinematic Representation

Although the D-H representation is used extensively there is a fundamental flaw with this method [1]. This problem arises from the all motions being defined in terms of z and x axes, and neglecting the y-axis. If there is any motion in the y-axis the D-H representation will fail. For example, if there is any deviation in parallel joints there will be some motion required in the related y-axis. These deviations can occur in robots due to inaccuracies in the manufacturing process. It is assumed in this course that there are not any such deviations and the D-H representation methodology is always valid.

References

[1] Niku, S.B., *Introduction to Robotics*, Pearson Education, 2001.

[2] Sciavicco, L. & Siciliano, B., *Modelling and Control of Robot Manipulators*, Springer, 2001.